

# Research Plan

Toward a Material Inference Prover:  
Mechanisms for Controlled Reasoning in Normative and Argumentative Domains

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## Abstract

In formal domains, theorem provers provide controlled reasoning through explicit rules and verifiable steps. In normative and argumentative domains (legal, moral, political reasoning), no such provers exist: reasoning relies on implicit premises, unstated reasoning strategies, and contested concepts. This thesis investigates the mechanisms required to extend controlled reasoning to these domains, working toward a *material inference prover*. We identify three classes of control mechanisms: (i) elicitation and structure, making implicit reasoning content formally explicit; (ii) systematic multi-step decomposition into typed functional components; and (iii) iterative refinement via locally verifiable quality proxies. These mechanisms are investigated across three domains: legal reasoning (emphasis on refinement and proxies), political and argumentative reasoning (emphasis on elicitation), and moral reasoning via reflective equilibrium (emphasis on decomposition and convergence). A cross-cutting question is how far the elicitation of reasoning structure should go, and where further formalization ceases to be productive.

## 1 Introduction

In formal domains (mathematics, logic, programming) we have *provers*: systems like Lean and Isabelle that verify whether a conclusion follows from premises, search for proofs, and suggest strategies. They work because reasoning in these domains follows explicit rules.

In many other domains, particularly argumentative, moral, and legal reasoning, this is less the case. While formal and, to a degree, scientific reasoning can rely on explicit rules and verifiable ground truth, these domains often cannot. When a court decides whether a state violated Article 8 of the European Convention on Human Rights, the inferences depend on *what the words mean*, not on logical form alone; in philosophy, this is called **material inference** [Brandon, 1994, Sellars, 1953]. Two features distinguish these domains:

1. **Implicit premises.** Material inferences rely on unstated assumptions. “Countries collect taxes → they steal from citizens” relies on the hidden premise that taking money without individual consent is theft. A different hidden premise (that redistribution reduces inequality) yields the opposite conclusion: “Countries collect taxes → they strive towards equality.” Arguments with such missing premises are called *enthymemes*, and they are the norm rather than the exception in real-world reasoning.
2. **Implicit reasoning strategies.** Correct reasoning in these domains still requires selecting the right information and connecting it through appropriate patterns, but the patterns are implicit rather than explicit. For instance, ECHR courts routinely apply *proportionality balancing*, weighing the severity of a rights interference against the importance of the aim pursued. Lawyers know how to do this, but how much weight to give each side cannot be reduced to formal logic.

Raw text (based on *Stoll v. Switzerland*, ECHR 2007)

The Court observes that the applicant's articles addressed a matter of public interest, namely the diplomatic strategy regarding Holocaust victim compensation. However, the articles presented the ambassador's position in a reductive and sensationalist manner, likely to mislead readers. The protection of diplomatic correspondence serves the legitimate aim of maintaining confidence in international negotiations. Having regard to the modest nature of the penalty imposed and the fact that it did not constitute prior restraint, the Court concludes that the interference was proportionate to the legitimate aim pursued.

⇓ *Elicitation: make implicit structure explicit*

Elicited structure

**Reasoning pattern:** Proportionality balancing (weighing interference against legitimate aim).

**Explicit premises:**

(P1) The articles addressed a matter of public interest.

(P2) The presentation was reductive and sensationalist.

(P3) Diplomatic correspondence requires confidentiality.

(P4) The penalty was modest and not a prior restraint.

**Implicit premises (surfaced):**

(P5) Sensationalist presentation weakens a public-interest justification.

(P6) A modest post-publication penalty is less restrictive than a ban.

**Argumentation scheme:** Argument from competing values (P1 vs. P3), resolved by assessing the journalist's conduct (P2, P5) and the severity of the penalty (P4, P6).

⇓ *Formal translation*

Formal translation

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(1) addresses(articles, public_interest)
(2) presentation(articles, reductive ∧ sensationalist)
(3) serves(protection_diplomatic_correspondence, legitimate_aim)
(4) modest(penalty) ∧ ¬prior_restraint(penalty)
(5) (1) ∧ (2) ∧ (3) ∧ (4) → proportionate(interference)
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Figure 1: Progressive elicitation of implicit structure from legal reasoning. The raw text (top) contains hidden premises and an unnamed reasoning pattern. Elicitation (middle) makes this structure explicit, enabling inspection and control. Full formalization (bottom) represents individual premises but cannot capture the normative weighing that connects them.

Normative reasoning is, however, not without structure. Philosophy and argumentation theory have identified much of it. To give a few examples: argumentation schemes describe recurring argument patterns, informal logic provides criteria for assessing argument quality, and reflective equilibrium formalizes coherence-seeking in normative reasoning. These resources have rarely been operationalized computationally. Current LLM-based reasoning can produce plausible arguments in these domains, but without control over which premises are assumed, which strategy is applied, or whether the result is structurally sound. This thesis investigates the mechanisms required to achieve such control.

Between fully implicit reasoning (as produced by unconstrained LLMs) and fully formal reasoning (as handled by theorem provers) lies a spectrum of possible levels of explicitness. One can progressively make the implicit structure of an argument more explicit, a process we call *elicitation*: surfacing hidden premises, naming reasoning patterns, normalizing vocabulary, identifying contested concepts, and decomposing the argument into verifiable steps. Each operation constrains the context and inference, bringing reasoning closer to a regime where it can be controlled and verified. Full formal translation (e.g., into first-order logic) can represent individual premises but struggles to capture the reasoning that connects them, particularly when that reasoning involves normative weighing or contested concepts. Figure 1 illustrates this progression on a passage of legal reasoning. How far along this spectrum one should go, and which operations matter most, is a central question of this thesis.

**This thesis investigates the mechanisms required for controlled LLM-based reasoning in normative and argumentative domains**, drawing on resources from argumentation theory and philosophy. The long-term vision is a *material inference prover*: a model that, like formal provers, can produce inferences, verify arguments, and reconstruct reasoning, but for normative domains where formal rules are largely absent. The central contribution is not the conclusions themselves but the *control* over how they are reached: each step is explicit, typed, and verifiable. We identify three classes of control mechanisms: (i) *elicitation and structure*, making implicit premises, reasoning patterns, and argumentative structure formally explicit; (ii) *systematic multi-step decomposition*, composing typed functional components for complex reasoning

tasks, where the composition itself constitutes a form of inferential control; and (iii) *iterative refinement via proxies*, using locally verifiable quality criteria as feedback signals for monotonic improvement. Where existing neuro-symbolic approaches pair LLMs with formal solvers, this thesis extends the paradigm to domains where no formal solver exists.

We formulate the following research questions and hypotheses.

#### Research Questions

**RQ1** *Elicitation and structure.* What are the mechanisms for eliciting and structuring implicit reasoning content in normative domains, and to what extent can they make premises, reasoning patterns, and argumentative structure formally explicit?

**RQ2** *Multi-step decomposition.* How can complex normative reasoning be decomposed into typed functional components (e.g., premise elicitation, entailment verification, consistency checking, contested concept identification), and what is the impact of this systematic decomposition on reasoning correctness and control?

**RQ3** *Iterative refinement.* Under what conditions do iterative refinement loops, grounded in locally verifiable quality proxies and LLM-based judges, converge toward improved reasoning in normative domains?

*Cutting across all three:*

**RQ4** *Degree of formalization.* To what degree can the implicit structure of normative reasoning be made formally explicit? Which structural elements matter most for achieving control, and where does further formalization cease to be productive?

#### Hypotheses

**H1.** Elicitation guided by argumentation-theoretic structure (schemes, critical questions, premise-conclusion decomposition) yields representations that enable effective verification and control of normative reasoning.

**H2.** Systematic decomposition of normative reasoning into typed functional components (each with a specific inferential role) improves correctness and controllability compared to monolithic generation.

**H3.** Iterative refinement driven by locally verifiable quality proxies (such as premise completeness, scheme compliance, and coherence metrics) produces monotonic improvement in reasoning quality, with convergence properties analogous to those demonstrated in formal domains.

**H4.** Partial elicitation of reasoning structure already yields meaningful control. Full formalization into logical form is neither necessary nor always possible for normative reasoning, as it risks losing information that drives the inference.

This work is situated within the SNSF project M-RATIONAL [M-RATIONAL, 2024], which develops multi-perspective reasoning for argumentative AI.

**Keywords:** material inference, controlled reasoning, argumentation schemes, quality proxies, iterative refinement, agentic reasoning models, neuro-symbolic AI

## 2 State of the Art

### 2.1 From formal to normative reasoning

Table 1 frames the thesis through a spectrum of reasoning domains.

**Formal reasoning** has explicit rules and unambiguous ground truth. Theorem provers like Lean [de Moura and Ullrich, 2021] and Isabelle [Paulson, 1994] use *tactics*, modular reasoning strategies that can be selected and composed to construct proofs. Even here, choosing the right strategy matters enormously.

**Scientific reasoning** still has ground truth (the external world constrains what is correct), but the reasoning process is more complex: relevant information must be extracted from heterogeneous sources, and inference rules are not always fully formalized. Among the approaches that have been developed for this

| Domain                      | Rules explicit?  | Ground truth? | Existing approaches                  |
|-----------------------------|------------------|---------------|--------------------------------------|
| Formal (math, logic)        | Yes              | Yes           | Lean, Isabelle, Z3                   |
| Scientific                  | Partially        | Mostly yes    | Neuro-symbolic models (among others) |
| Legal, moral, argumentative | Largely implicit | Less clear    | <b>Gap (this thesis)</b>             |

Table 1: The reasoning spectrum. This thesis targets the rightmost column.

setting, neuro-symbolic methods [Garcéz et al., 2019, Olausson et al., 2023, Pan et al., 2023, Xu et al., 2025] combine LLM flexibility with symbolic solvers. A common assumption across these approaches, however, is that a formal solver exists for the target domain.

**Normative and argumentative domains** (legal, moral, political reasoning) involve judgments about values, norms, and what ought to be done. They have no such solvers, and it is far less clear how existing logical solvers could be applied. Reasoning in these domains depends on implicit premises and implicit reasoning strategies. Ground truth, in the scientific sense, is often absent: what counts as correct reasoning is less about converging on a single answer and more about reaching a well-justified balance among competing considerations [Queloz, 2025].

## 2.2 Mechanisms of control: elicitation, decomposition, refinement

The thesis investigates three classes of mechanisms for controlling LLM-based reasoning in normative domains. We organize the literature accordingly.

### 2.2.1 Elicitation and structure

A growing body of work addresses the elicitation of implicit structure from arguments. On the premise side, Chakrabarty et al. [2021] generate missing premises in enthymemes using a seq2seq model fine-tuned on abductive reasoning data. Gupta et al. [2024] show that prompting LLMs with Toulmin’s argumentation model, which decomposes arguments into claim, reason, and *warrant* (the implicit principle connecting them), enables zero-shot premise recovery, outperforming theory-agnostic prompts. Ku et al. [2025] use multi-agent LLM debate, with adversarial exchange improving accuracy over single-agent approaches. Delas et al. [2024] propose the first framework that uses argumentation scheme classification to guide premise recovery, identifying *what kind* of premise is missing based on the argument’s structure.

On the reasoning pattern side, Ruiz-Dolz et al. [2025] create a corpus of 441 arguments from BBC political dialogue annotated with 24 Walton schemes, achieving classification F1 up to 0.71. Bezou-Vrakatseli et al. [2024] build the Ethix dataset of 686 ethical arguments annotated with 8 schemes. Bezou-Vrakatseli et al. [2025] evaluate whether LLMs can identify schemes zero-shot, finding that larger models succeed but that missing premises significantly increase difficulty, confirming the coupling between premises and reasoning patterns. Stahl et al. [2023] provide a 40K-instance dataset for enthymeme detection.

### 2.2.2 Multi-step decomposition and agentic control

Several lines of work decompose complex reasoning into specialized sub-tasks. Jin et al. [2022] introduce MoralCoT, a chain-of-thought strategy from contractualist moral philosophy that decomposes moral reasoning into typed steps (rule identification, purpose reflection, utility assessment), outperforming baselines by 6.2% F1. Stahl et al. [2025] fine-tune LLMs on 52K argumentation-specific instructions spanning multiple reasoning sub-tasks.

In the neuro-symbolic space, Logic-Explainer [Quan et al., 2024] decomposes ethical reasoning into generation, autoformalization, verification, and refinement stages, improving logical validity from 22% to 55%. Quan et al. [2025a] extend this with Faithful-Refiner, using Isabelle for formal step-by-step

verification. [Quan et al. \[2025b\]](#) introduce recursive decomposition with local diagnostic-guided refinement, achieving 21–49% improvements in verification rates. [Mohr et al. \[2025\]](#) decompose text-to-SQL generation into typed stages with localized violation detection, demonstrating that stage-specific control outperforms monolithic generation. These approaches demonstrate the value of decomposition, but rely on formal solvers not available for normative reasoning. Reflective equilibrium [[Beisbart et al., 2021](#)] offers a contrasting example: a reasoning procedure that naturally decomposes into typed steps (theory proposal, commitment revision, coherence assessment) without requiring formal solvers, but remains almost entirely unexplored computationally [[Awad et al., 2022](#), [Ma et al., 2025](#)].

### 2.2.3 Iterative refinement via proxies

When no formal verifier exists, alternative feedback mechanisms are needed for refinement. [Zhang et al. \[2025\]](#) propose epistemic ensembles of LLM-based judges, decomposing quality into locally verifiable dimensions (logical preservation, mathematical consistency, formal validity, formal quality) and demonstrating that such ensembles correlate with human assessment better than holistic metrics. [Alshomary et al. \[2022\]](#) generate arguments controlled by explicit moral foundations, showing that making value commitments explicit improves argument quality. Evaluations of LLM reasoning in argumentative domains [[Chen et al., 2024](#), [Zhou et al., 2024](#)] consistently find partial capabilities but no control or structural guarantees, with moral reasoning consistency heavily inflated by position bias, underscoring the need for controlled refinement mechanisms.

The field of informal logic has developed quality criteria that could serve as proxies for normative reasoning [[Groarke, 2026](#)], most notably the **ARS criteria**: *acceptability*, *relevance*, and *sufficiency* of premises [[Johnson and Blair, 1977](#)]. For moral reasoning, [Beisbart et al. \[2021\]](#) formalize structural properties of reflective equilibrium (*account*, *systematicity*, *faithfulness*) as measurable quality metrics. Operationalizing such criteria as automatic proxies for iterative refinement is a key open challenge [[Haas et al., 2026](#)].

### 2.2.4 Gaps

Three gaps emerge from this survey.

1. **No principled account of elicitation for normative reasoning.** Existing work makes various aspects of reasoning structure explicit (e.g., Toulmin warrants, argumentation schemes, moral foundations), but each approach picks a particular level of elicitation without systematically investigating which structural elements matter most for control, what the mechanisms of elicitation are, or under what conditions progressive elicitation converges toward reliable reasoning.
2. **Quality proxies for normative reasoning are undeveloped.** Locally verifiable properties of the reasoning process, informed by argumentation theory and informal logic, have not been operationalized as automatic feedback mechanisms for iterative refinement.
3. **Epistemically informed reasoning procedures are unexplored.** Existing neuro-symbolic models rely on formal solvers (Prolog, Isabelle) not available for normative domains. Reasoning procedures that provide structure without requiring formal logic, such as reflective equilibrium, remain almost entirely unexplored computationally.

## 3 Research Plan

### 3.1 Control mechanisms

The thesis proposes and evaluates mechanisms for building a *material inference prover* for normative and argumentative domains. Like formal provers, which can verify proofs, search for proofs, and suggest

strategies, the material inference prover is designed to produce inferences, verify existing arguments, and reconstruct reasoning in normative domains. The central value is control: reasoning proceeds through explicit, typed, and verifiable steps rather than opaque generation. The model is organized around three classes of control mechanisms (corresponding to RQ1–RQ3), designed to be adaptable across domains of analysis. RQ4 (degree of formalization) cuts across all three.

The model comprises typed functional components, each materializing a specific notion of inferential control:

- Components that *elicit* implicit content: surfacing hidden premises, identifying enthymemes, naming reasoning patterns, flagging contested concepts. (RQ1)
- The *systematic decomposition and orchestration* of these components: how to decompose complex reasoning into specialized sub-tasks, select the appropriate component, and route between elicitation and refinement. The composition itself constitutes a form of inferential control. (RQ2)
- Components that *verify and refine* reasoning through iterative feedback: checking entailment, assessing consistency, evaluating scheme compliance, and using locally verifiable quality proxies as control signals for monotonic improvement, analogous to the role of theorem provers in formal domains. (RQ3)

These components interact through iterative refinement loops: elicitation produces structured representations, verification identifies weaknesses, and refinement feeds diagnostic information back into elicitation. The quality proxies that drive refinement are grounded in argumentation theory and informal logic (e.g., premise completeness, scheme compliance, the ARS criteria [Johnson and Blair, 1977, Groarke, 2026]; for moral reasoning, the RE metrics of account, systematicity, and faithfulness [Beisbart et al., 2021]).

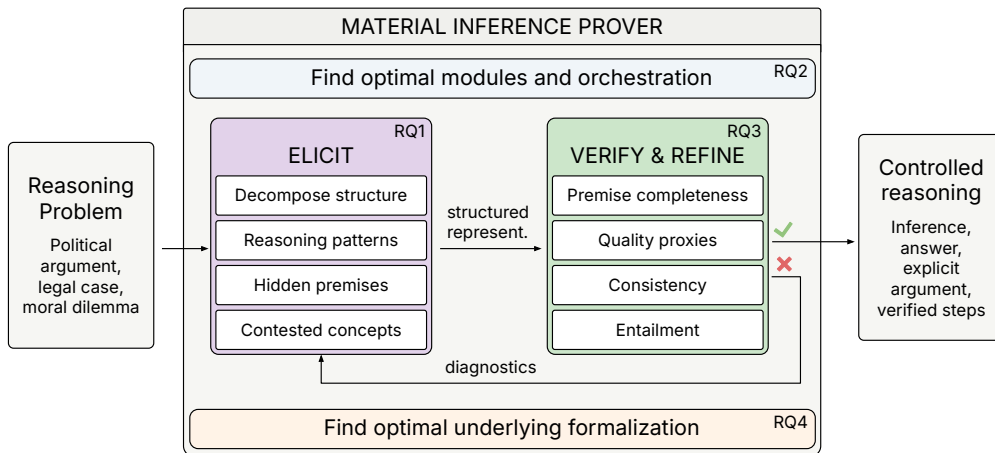


Figure 2: Architecture of the material inference prover. Typed functional components for elicitation (RQ1) and verification/refinement (RQ3) interact in an iterative loop. The systematic decomposition and orchestration of these components is itself a mechanism of control (RQ2). Quality proxies provide the feedback signals that drive convergence. RQ4 (degree of formalization) spans the full model.

### 3.2 Domains of analysis

The mechanisms described above are investigated across three domains, each selected because it foregrounds a different aspect of the control problem. The selection follows two criteria: (i) the availability of established data and evaluation resources, and (ii) coverage of different challenges for the control mechanisms.



### 3.2.1 Legal reasoning (emphasis: iterative refinement and proxies)

*To be refined as the thesis develops.*

Legal reasoning, and ECHR case law in particular, is a domain where judges are trained to spell out their reasoning, cite precedents, and name the standards they apply. This relative explicitness makes it well suited for developing and validating the quality proxies that drive iterative refinement (RQ3), since expert reasoning provides a reference against which proxies can be calibrated. Legal reasoning also foregrounds specific inferential types, including entailment from precedent, proportionality balancing, and the interpretation of contested legal concepts (e.g., “proportionate,” “necessary in a democratic society”). The model is instantiated with typed functional components for each of these inferential types, and the iterative refinement loop is driven by quality proxies including premise completeness, contested concept identification, scheme compliance [Walton et al., 2008], and vocabulary precision.

**Data:** The ECtHR dataset [Chalkidis et al., 2021] (11,000 cases with expert rationales), LAR-ECHR [Chlapanis et al., 2024] (argument chain annotations), and LegalBench [Guha et al., 2023] (162 legal reasoning tasks).

### 3.2.2 Political and argumentative reasoning (emphasis: elicitation)

*To be refined as the thesis develops.*

In political and argumentative discourse, premises are frequently left unstated, contested concepts (e.g., “freedom,” “responsibility”) are used without disambiguation, and competing frameworks are assumed rather than acknowledged. This domain is well suited for studying elicitation (RQ1) and the degree of formalization (RQ4): how much implicit structure can the model’s elicitation mechanisms recover, and at what level does the recovered structure become useful for verification and refinement? Making the structure of political arguments transparent also has potential value for democratic deliberation. The base mechanisms involve working with argumentation schemes [Walton et al., 2008] as the primary vocabulary for typing reasoning steps, and with the ARS criteria [Johnson and Blair, 1977] as proxies for argument quality.

**Data:** The QT-Schemes corpus [Ruiz-Dolz et al., 2025] (441 arguments from BBC political dialogue, annotated with 24 Walton schemes), the EthiX dataset [Bezou-Vrakatseli et al., 2024] (686 ethical arguments annotated with 8 schemes), and the Kialo debate platform [Kialo, 2024].

### 3.2.3 Moral reasoning (emphasis: decomposition and convergence)

*Research currently underway.*

Moral reasoning requires not only the mechanisms applicable to argumentation in general but also coherence-seeking across principles and particular judgments. This makes it well suited for studying decomposition (RQ2): how to break complex reasoning into typed functional components whose systematic composition achieves convergence. The model is instantiated with components for theory proposal, commitment revision, and coherence assessment, drawing on the full range of control mechanisms (RQ1, RQ3) and raising the question of how far formalization should go (RQ4).

The specific reasoning procedure investigated is reflective equilibrium (RE) [Rawls, 1971], the central methodology of moral philosophy. The basic idea is that one starts from case judgments and proposes principles that account for them. When a principle conflicts with a judgment, either the principle is revised or the judgment is reconsidered, and the process repeats until they cohere. For example, given the judgments “lying to protect a friend is permissible” and “lying to a patient about a diagnosis is wrong,” one might propose “lying is permissible when it prevents harm.” But this conflicts with the second judgment (lying to a patient might prevent emotional harm), so either the principle is revised (e.g., adding an autonomy condition) or the judgment is reconsidered. Beisbart et al. [2021] formalize this process

into computational steps with quality metrics (*account*, *systematicity*, *faithfulness*; see Figure 3), yet RE remains almost entirely unexplored in AI [Awad et al., 2022, Ma et al., 2025].

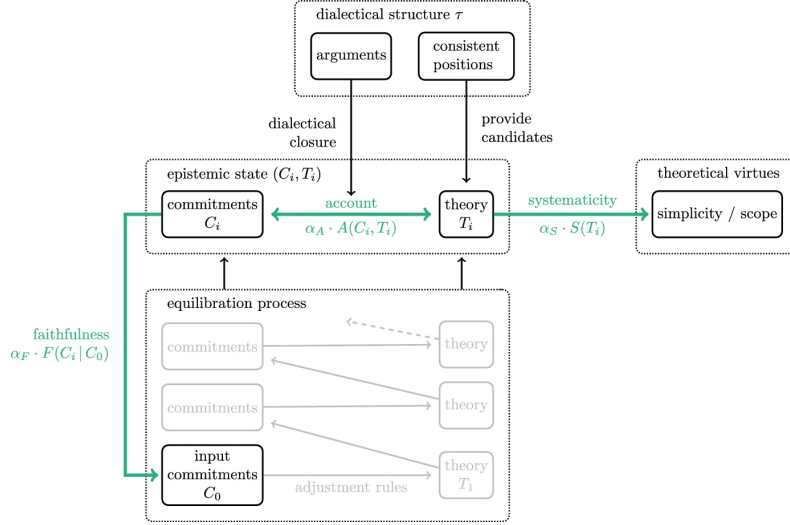


Figure 3: Formal model of reflective equilibrium (from Freivogel [2023]). The epistemic state (commitments  $C_i$  and theory  $T_i$ ) is subject to three desiderata: account, systematicity, and faithfulness. Adjustment rules alternate between revising the theory and revising the commitments until a fixed point is reached. The sequence of revisions constitutes the reasoning trace.

**Task definition.** *Input:* a set of moral case judgments. *Output:* a set of principles and revised judgments in reflective equilibrium, together with the reasoning trace (the full sequence of revisions). This trace constitutes the moral reasoning itself and amounts to a justification of the final state reached.

**Approach.** We investigate in three stages:

1. **Assess RE operations in isolation.** RE decomposes moral reasoning into distinct operations: abduction upward (given case judgments, propose a principle), deduction downward (given a principle, derive implications or adjust commitments), and coherence assessment (given two configurations, identify which is better). We build targeted evaluations of each. Prior work suggests specific bottlenecks: Cheng et al. [2024] find that LLMs extract general rules from examples more reliably than they revise those rules when confronted with counterevidence, and Wilie et al. [2024] show that belief revision is difficult for LLMs.
2. **Implement the full RE model.** A reasoning model with typed functional components: one for theory proposal, one for commitment revision, and one for coherence assessment via quality proxies (account, systematicity, faithfulness). Does it converge? What principles does it find? What does it revise and why?
3. **Evaluate.** Does RE-guided reasoning improve structural quality compared to unconstrained LLM generation, chain-of-thought, MoralCoT [Jin et al., 2022], and single critique-and-revise [Ma et al., 2025]?

**Data:** MoReBench [Chiu et al., 2025] provides expert rubrics specifying which premises and reasoning steps should appear, a rare source of expert-defined quality criteria for the reasoning process.

### 3.3 Evaluation methodology

Evaluation addresses two dimensions: the quality of the reasoning produced and the effectiveness of the control mechanisms.



**Quality proxies.** Reasoning quality is assessed through locally verifiable proxies [Zhang et al., 2025] rather than holistic judgments. For general argumentation, these include premise completeness, scheme compliance, and the ARS criteria [Johnson and Blair, 1977]. For legal reasoning, they additionally include contested concept identification and vocabulary precision. For moral reasoning, Beisbart et al.’s RE metrics (*account, systematicity, faithfulness*) provide domain-specific quality measures. A cross-cutting question is whether these proxies generalize across domains.

**Baselines.** To ensure that findings are transferable across the current spectrum of models, evaluation is conducted across multiple model families and scales (both open and closed-source). Baselines include: unconstrained generation, chain-of-thought prompting, self-refinement [Ma et al., 2025], MoralCoT [Jin et al., 2022], and existing controlled reasoning approaches such as Logic-Explainer [Quan et al., 2024] and Faithful-Refiner [Quan et al., 2025a].

**Datasets.** For moral reasoning: MoReBench [Chiu et al., 2025] (expert rubrics on reasoning processes) and the ETHICS benchmark [Hendrycks et al., 2021] (justice, deontology, virtue, utilitarianism, commonsense morality). For legal reasoning: the ECtHR dataset [Chalkidis et al., 2021] (11,000 cases with expert rationales), LAR-ECHR [Chlapanis et al., 2024] (argument chain annotations), and LegalBench [Guha et al., 2023] (162 legal reasoning tasks). For argumentative reasoning: the QT-Schemes corpus [Ruiz-Dolz et al., 2025] (441 arguments, 24 Walton schemes), Ethix [Bezou-Vrakatseli et al., 2024] (686 ethical arguments, 8 schemes), OpenDebateEvidence [OpenDebateEvidence, 2024] (3.5M debate documents), ArgBench [ArgBench, 2025] (LLM argumentation benchmark), and the Kialo debate platform [Kialo, 2024].

**Evaluation questions.** Beyond comparing against baselines, the evaluation addresses several cross-cutting questions: (i) *mechanism contribution*: through ablation, does each mechanism (elicitation, decomposition, refinement) add value over the others? (ii) *convergence*: does the iterative refinement loop improve reasoning quality monotonically over iterations, and under what conditions does it converge? (iii) *generalizability*: do proxies and mechanisms developed in one domain transfer to others? (iv) *error analysis*: what types of failures occur (missing premises, incorrect scheme application, unresolved contested concepts), and which mechanisms address which failure types?

**Selection criteria.** The choice of domains, datasets, and baselines follows explicit eligibility criteria: domains are selected for coverage of different challenges for the control mechanisms and availability of expert-annotated resources; baselines are selected to represent the current state of the art across mechanism types and model scales.

## 4 Research Performed (Year 1)

Two contributions from the first year establish methods in formal and scientific reasoning that the thesis extends to normative and argumentative domains.

### 4.1 Adaptive neuro-symbolic reasoning (EACL 2025)

Our first contribution [Xu et al., 2025] addresses a limitation of existing neuro-symbolic models: they couple to a single formal solver at design time. We introduced an adaptive framework that (1) uses an LLM to identify the required formal reasoning strategy and (2) dynamically selects the appropriate solver.

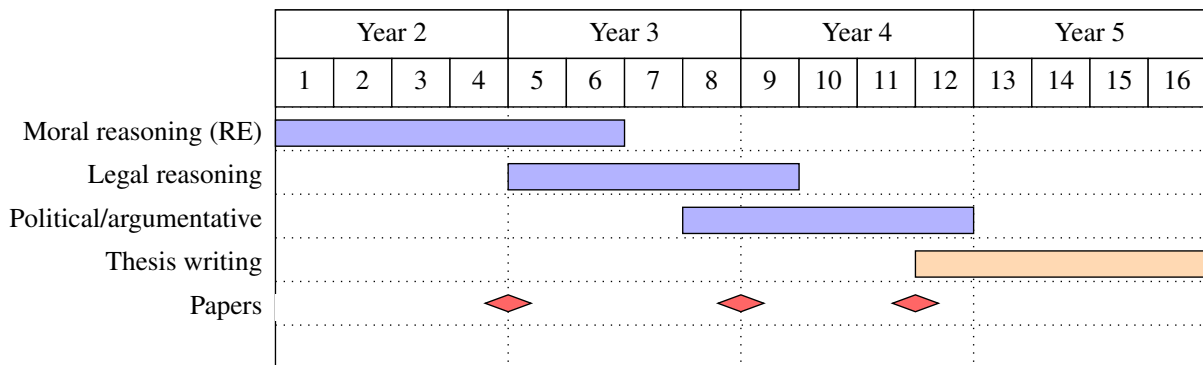
**Results:** LLMs predict formal reasoning strategies at >90% accuracy. The adaptive framework outperforms GPT-4o by 17% and DeepSeek-V3.1 by 6%. Even pure LLM methods improve with adaptive strategy selection (+10% zero-shot GPT-4o).

## 4.2 Scientific reasoning benchmark (in submission)

Our second contribution is a benchmark that disentangles two stages of reasoning: (i) *information extraction*, identifying and structuring premises from messy input, and (ii) *principled inference*, applying rules to the structured representation. It spans deductive, inductive, and causal tasks, each systematically varied along both axes. A *problem obfuscation scheme* converts clean logical problems into realistic scientific documents (lab notes, database entries, paper extracts), testing extraction under increasing difficulty.

**Results:** Direct LLM performance degrades sharply as inference complexity increases: on deduction, the best LLM drops from 84% (easy) to 58% (hard); on causal reasoning, from 97% to 75%. Obfuscation further reduces performance, with drops of up to 39 percentage points. Both inference complexity and realistic presentation pose significant challenges for LLMs, and the two compound.

## 5 Schedule



Each quarter corresponds to approximately 3 months. Moral reasoning (RE) is the current focus and drives the initial development of the control mechanisms. The political/argumentative base case and legal reasoning follow, each inheriting and extending these mechanisms to domains of increasing structural explicitness. The final quarters are reserved for thesis writing, though individual chapters are drafted alongside each part.

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